Seat No. _____ Enrollment No. _____

THE CHARUTAR VIDYA MANDAL UNIVERSITY

B.E.– SEMESTER 6

AUGUST 2023 (BACKLOG) EXAMINATION

Course Title: Smart Cities Planning And Management

Course Code: 102030622

Total Printed Pages: 02

Date: 08/08/2023 **Time: 10.00 am to 12.30 pm** **Maximum Marks: 60**

Instructions:

- Attempt all questions.
- Numbers to the right indicate full marks for each question.
- Make suitable assumption if required, do specify the same.

Q: 1 **Answer the Following. (Each question carries one marks).** **[10]**

(i) Define Smart city.

(ii) Smart Cities Mission comprising 100 Indian Cities was launched on _____.
a) 25th June 2015 b) 10th August 2014 c) 1st December 2017 d) 10th Nov.2020

(iii) Write the full form of ITS.

(iv) Which of the following Green buildings rating systems working in India?
a)LEED b) GRIHA c) Both a) and b) d) None of the above

(v) Green Building practice includes
a) Energy efficiency b) Recycled materials c) Environmental protection d) All of the above

(vi) What is the full form of IGBC?

(vii) Which technologies are used in ITS?
a)Communication technology b) Artificial vision system
c) Digital Mapping d) All of the above

(viii) In which state "Rani ki vav" step well is located for water conservation?
a)Gujarat b) Punjab c) Rajasthan d) Haryana

(ix) The smart gird will help consumers to manage their electricity bills. a) True b) False

(x) Smart grid technologies are aimed at improvement of
a)only power transmission system b) only power distribution system
c) both a and b d) None of the above

Q: 2 **(A)** Explain the concept of Smart city. What the requirements of smart cities? **[04]**

(B) Discuss the advantages of Smart city. **[06]**

Q: 3 **(A)** Discuss any case study of Smart city. **[05]**

(B) Write a note on : Intelligent Transport System **[05]**

OR

(B) What is Smart Grid? Discuss characteristics of Smart Grid **[05]**

Q: 4 (A) Describe affordable housing in smart cities. [05]

(B) What is green building? Explain concept of green building. [05]

OR

(B) Discuss the key technologies needed for the Smart City development. [05]

Q: 5 (A) Write a note on Fuels for Future. [05]

(B) Describe various applications of GIS in transportation engineering. [05]

OR

(B) Discuss various phases of Smart City project management. [05]

Q: 6 (A) What is smart city challenge? Give procedure for selection of smart city. [05]

(B) Describe various smart water management technologies. [05]

OR

(B) Write a note on : 1) Early Warning system 2) Smart vehicles [05]

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